EVERY SCIENTIST NEEDS TO KNOW HOW TO CODE

AI EDITION

Tony Rice, NASA Ambassador

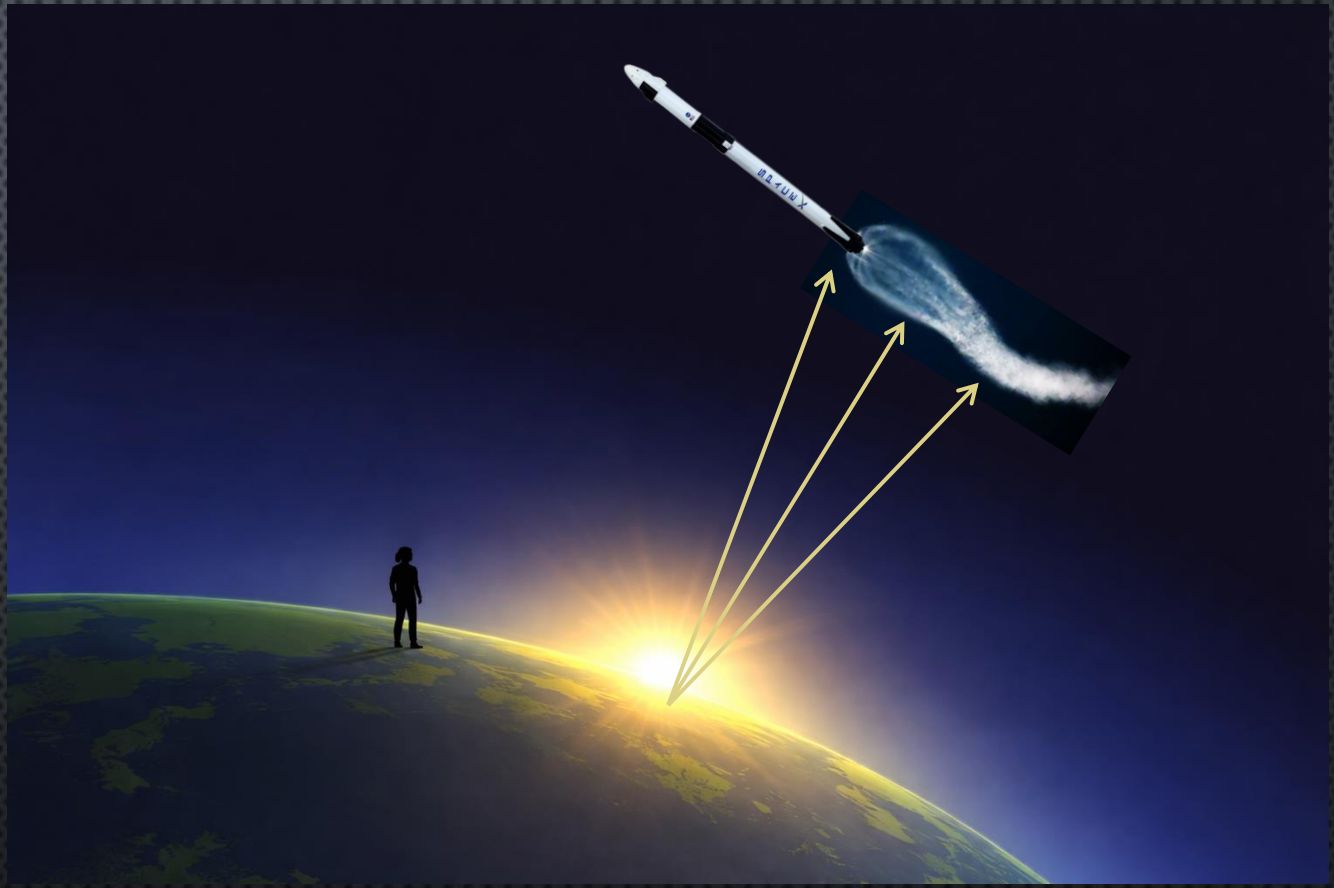
Garner, NC, July 2025
Esai Williams/WRAL

Palm Coast, FL, Oct 2025
Julio Roman/ Spectrum News

Deming, NM,

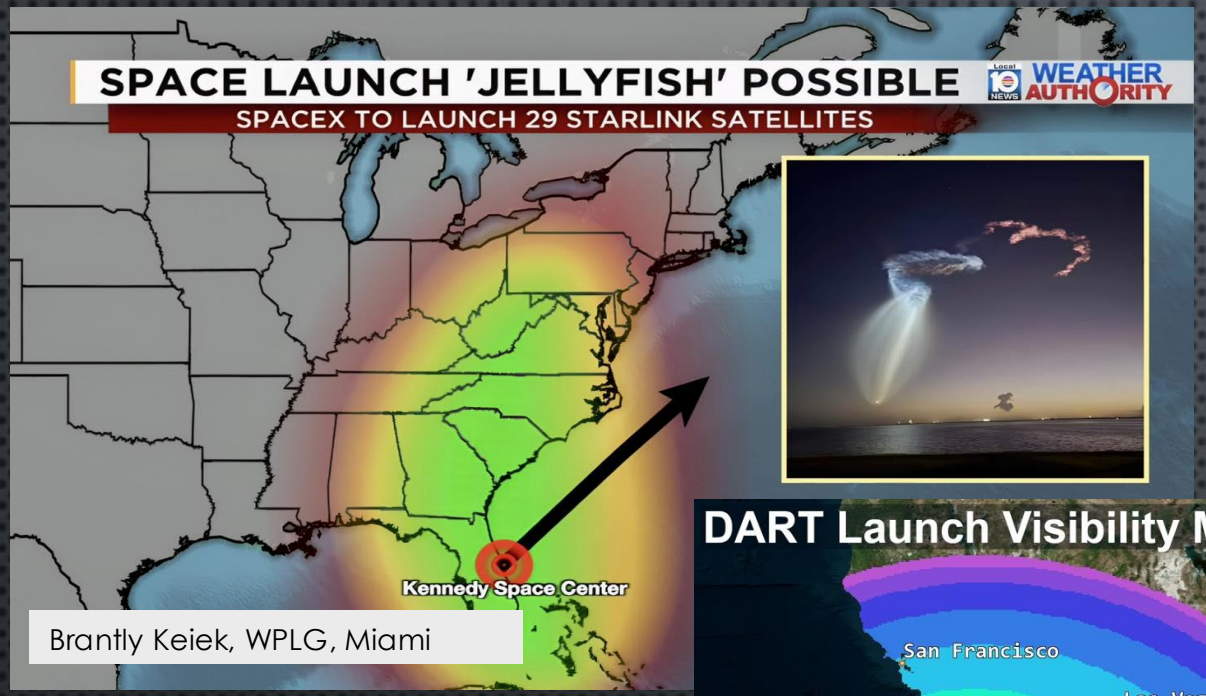
GEOMETRY OF THE SPACE JELLYFISH

- **Observer is in darkness, rocket is lit**
- **Low solar angle:** sun elevation around -10° (~1 hour before sunrise / after sunset)
- **Plume expands as air pressure decreases**

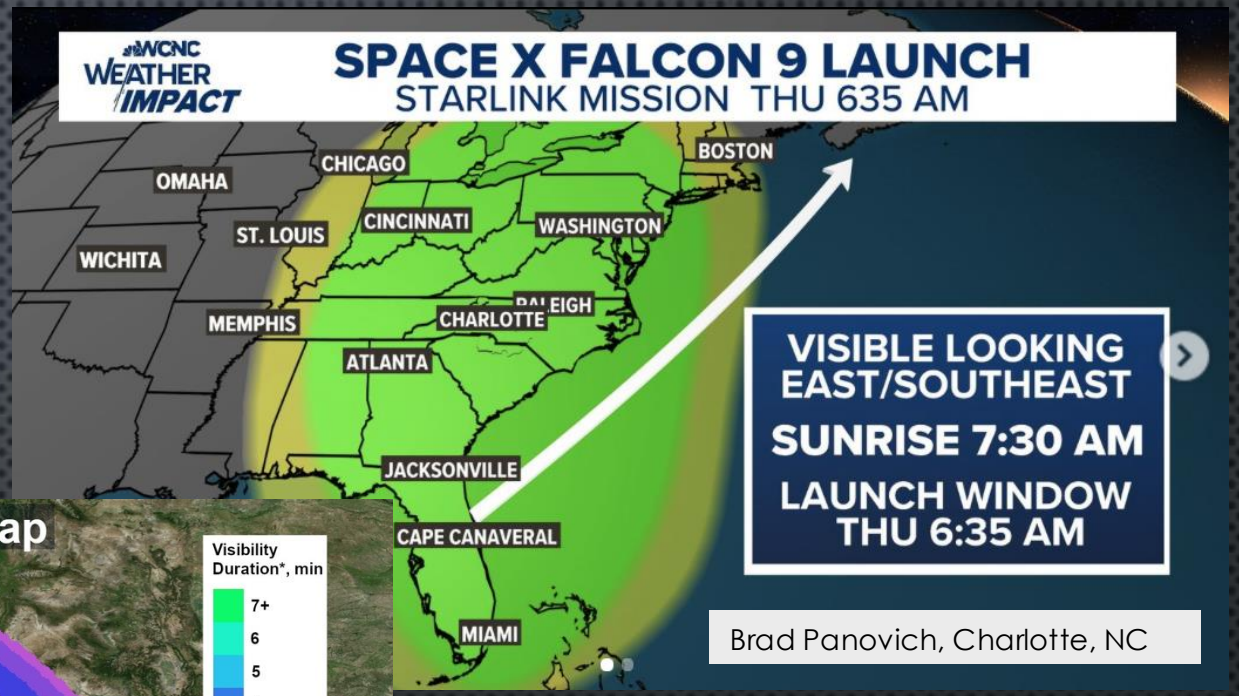


1. draw a diagram showing why rockets plume is lit by the sun during launches around twilight while its dark on the ground for observers
2. Remove the lines, and labels, move the sun from the sky to rising over the horizon

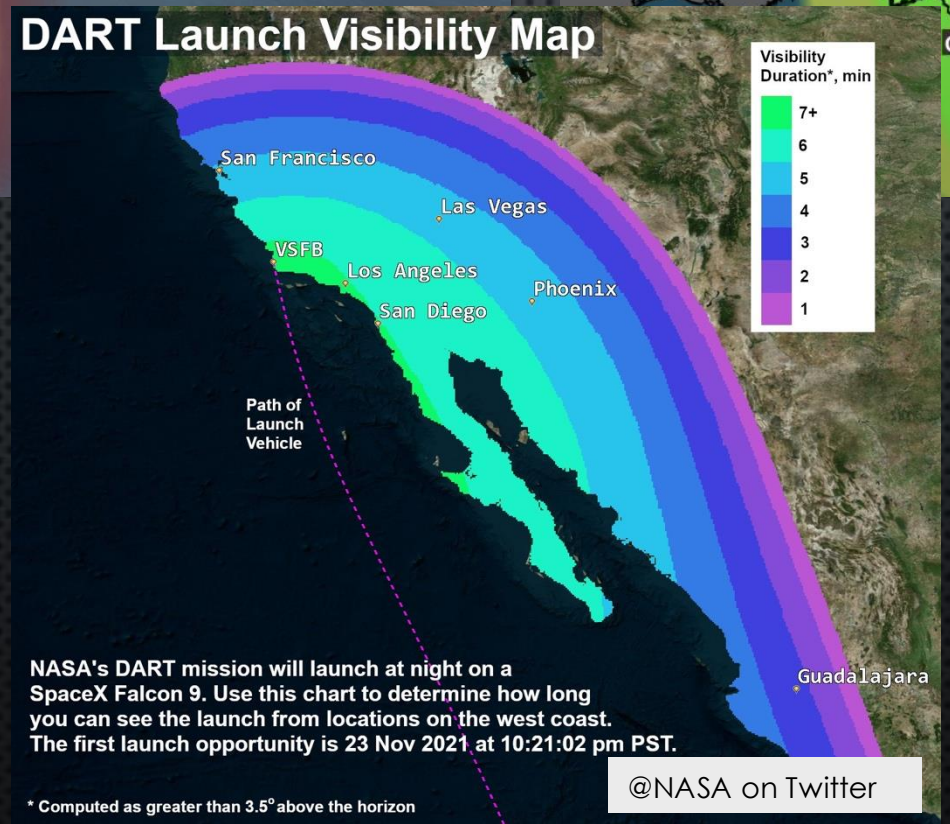
IDEAS ON HOW TO COMMUNICATE



Brantly Keiek, WPLG, Miami



Brad Panovich, Charlotte, NC



@NASA on Twitter

SPACE JELLYFISH RECIPE

1. **CANDIDATE LAUNCHES**
2. LAUNCH TIME AROUND 1 HOUR BEFORE SUNRISE OR AFTER SUNSET
   - COORDINATES OF LAUNCH SITE
   - SUN ALTITUDE: -12° TO -8°
3. TRAJECTORY ~~WITHIN VIEW OF OBSERVER~~ AT LEAST 3.5° ABOVE THE HORIZON
   - OBSERVERS COORDINATES

PROMPT:
Create a python function that finds upcoming launches from a launch site in a given state, defaulted to Florida. User the endpoint https://fdo.rocketlaunch.live/json/launches/next/5 as a data source.

RESPONSE:

```
def get_upcoming_launches(state="FL"):
    url = "https://fdo.rocketlaunch.live/json/launches/next/5"

    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()
        data = response.json()
    except requests.RequestException as e:
        print(f"Error fetching launch data: {e}")
        return []

    launches = data.get("result", [])
    filtered_launches = []

    for launch in launches:
        location = launch.get("pad", {}).get("location", {})
        state_name = location.get("state", "")

        if state_name and state.lower() in state_name.lower():
            filtered_launches.append(launch)

    return filtered_launches
```

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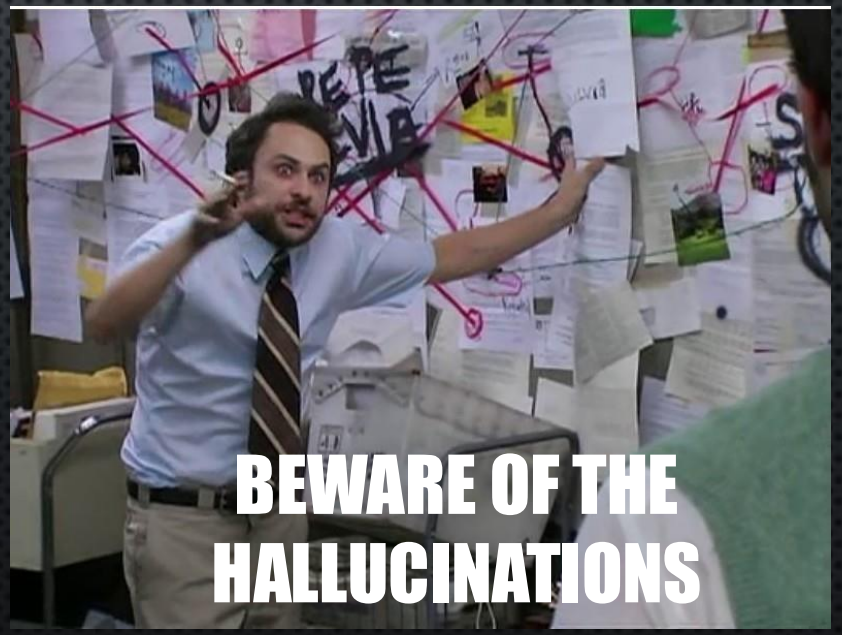
PROMPT:
Update that to also include the coordinates of the launch site

RESPONSE:
Some nonsense not worth showing here assuming longitude and latitude were in the data returned by the launch date API, they are not.



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   - OBSERVERS COORDINATES

PROMPT:
create a python function which looks up the coordinates of a location by name, return the longitude and latitude

RESPONSE:

```
def get_coordinates(location_name):
    url = "https://nominatim.openstreetmap.org/search"
    params = {
        "q": location_name,
        "format": "json",
        "limit": 1,  # return only the top result
    }

    response = requests.get(
        url, params=params, headers={"User-Agent":
"PythonGeocoder/1.0"}, timeout=10
    )
    response.raise_for_status()
    data = response.json()

    if data:
        lat = float(data[0]["lat"])
        lon = float(data[0]["lon"])
        return lat, lon
    else:
        return None
```

```
{'cospar_id': '',
 'date_str': 'Mar 27',
 'est_date': {'day': 26, 'month': 3, 'quarter': None, 'year': 2026},
 'id': 5923,
 'launch_description': 'A SpaceX Falcon 9 rocket will launch the Starlink '
                       '(10-44) mission on Friday, March 27, 2026 at '
                       '11:00\u202fAM (UTC).',
 'media': [],
 'mission_description': None,
 'missions': [{'description': None, 'id': 8355, 'name': 'Starlink (10-44)'}],
 'modified': '2026-03-26T13:43:32+00:00',
 'name': 'Starlink (10-44)',
 'pad': {'id': 3,
         'location': {'country': 'United States',
                      'id': 62,
                      'latitude': 28.4665327,
                      'longitude': -80.5631981,
                      'name': 'Cape Canaveral SFS',
                      'slug': 'cape-canaveral-sfs',
                      'state': 'FL',
                      'statename': 'Florida'},
         'name': 'SLC-40'},
 'provider': {'id': 1, 'name': 'SpaceX', 'slug': 'spacex'},
 'quicktext': 'Falcon 9 - Starlink (10-44) - Mar 27 (estimated) - '
              'https://rocketlaunch.live/launch/starlink-10-44 for info/stream',
 'result': -1,
 'slug': 'starlink-10-44',
 'sort_date': '1774609200',
 'suborbital': False,
 't0': '2026-03-27T11:00Z',
 'tags': [{'id': 109, 'text': 'B1067'}],
 'vehicle': {'company_id': 1, 'id': 1, 'name': 'Falcon 9', 'slug': 'falcon-9'},
 'weather_condition': 'Mainly Clear',
 'weather_icon': 'wi-day-sunny',
 'weather_summary': 'Mainly Clear\nTemp: 63\nWind: 1\n',
 'weather_temp': '62.90',
 'weather_updated': '2026-03-26T10:00:49+00:00',
 'weather_wind_mph': '1.40',
 'win_close': None,
 'win_open': None}
```

WHAT THE NUMBER OF DIGITS IN YOUR COORDINATES MEANS

LAT/LON PRECISION	MEANING
28°N, 80°W	YOU'RE PROBABLY DOING SOMETHING SPACE-RELATED
28.5°N, 80.6°W	YOU'RE POINTING OUT A SPECIFIC CITY
28.52°N, 80.68°W	YOU'RE POINTING OUT A NEIGHBORHOOD
28.523°N, 80.683°W	YOU'RE POINTING OUT A SPECIFIC SUBURBAN CUL-DE-SAC
28.5234°N, 80.6830°W	YOU'RE POINTING TO A PARTICULAR CORNER OF A HOUSE
28.52345°N, 80.68309°W	YOU'RE POINTING TO A SPECIFIC PERSON IN A ROOM, BUT SINCE YOU DIDN'T INCLUDE DATUM INFORMATION, WE CAN'T TELL WHO
28.52345 71°N, 80.68309 41°W	YOU'RE POINTING TO WALDO ON A PAGE
28.523457182°N, 80.683094159°W	"HEY, CHECK OUT THIS SPECIFIC SAND GRAIN!"
28.523457182818284°N, 80.683094159265358°W	EITHER YOU'RE HANDING OUT RAW FLOATING POINT VARIABLES, OR YOU'VE BUILT A DATABASE TO TRACK INDIVIDUAL ATOMS. IN EITHER CASE, PLEASE STOP.

https://xkcd.com/2170/

SPACE JELLYFISH RECIPE

1. Candidate launches
2. Launch time around 1 hour before sunrise or after sunset
   - Coordinates of launch site
   - Sun altitude: -12° to -8°
3. Trajectory ~~within view of observer~~ at least 3.5° above the horizon
   - Observers Coordinates

PROMPT:
add a function that returns the sun altitude in degrees at a given time, from a given longitude and latitude

RESPONSE:

```
def get_coordinates(location_name):
    url = "https://nominatim.openstreetmap.org/search"
    params = {
        "q": location_name,
        "format": "json",
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    response = requests.get(
        url, params=params, headers={"User-Agent":
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    if data:
        lat = float(data[0]["lat"])
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```

SPACE JELLYFISH RECIPE

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2. Launch time around 1 hour before sunrise or after sunset
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3. Trajectory ~~within view of observer~~ at least 3.5° above the horizon
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RESPONSE:

```
from astral import sun

def get_sun_altitude(lat, lon, iso_datetime):
    try:
        dt_utc = datetime.fromisoformat(iso_datetime.)
    except ValueError as e:
        print(f"Invalid ISO 8601 datetime: {e}")
        return None

location = LocationInfo(timezone="UTC", latitude=lat, longitude=lon)

    # Compute solar elevation (altitude)
    altitude = sun.elevation(observer=location.observer, dateandtime=dt_utc)
    return altitude
```

SPACE JELLYFISH RECIPE

PROMPT:
complete the script which listing upcoming launches, including the sun's altitude in the output and noting when that altitude is between -12 and -8

1. ~~Candidate launches~~
2. **Launch time around 1 hour before sunrise or after sunset**
   - ~~Coordinates of launch site~~
   - **Sun altitude: -12° to -8°**
3. ~~Trajectory within view of observer at least 3.5° above the horizon~~
   - ~~Observers Coordinates~~

RESPONSE:

```
if __name__ == "__main__":

    fl_launches = get_upcoming_launches()  # defaults to Florida
    for launch in fl_launches:
        print(f"{launch['name']} ({launch['vehicle']})")
        print(f"Provider: {launch['provider']}")
        print(f"When (UTC): {launch['t0']}")
        print(f"Where: {launch['pad']['location']['name']}")
        pad_coordinates = launch['pad']['location']
        sun_alt = get_sun_altitude(pad_coordinates['latitude'],
pad_coordinates 'longitude'], launch['t0'])
        print(f"Sun altitude: {sun_alt:.1f}", end ="")
        if -12 <= sun_alt <= -8:
            print(f"Space Jellyfish possible!")
        else:
            print()
        print("-" * 40)
```

MORE PROMPTS

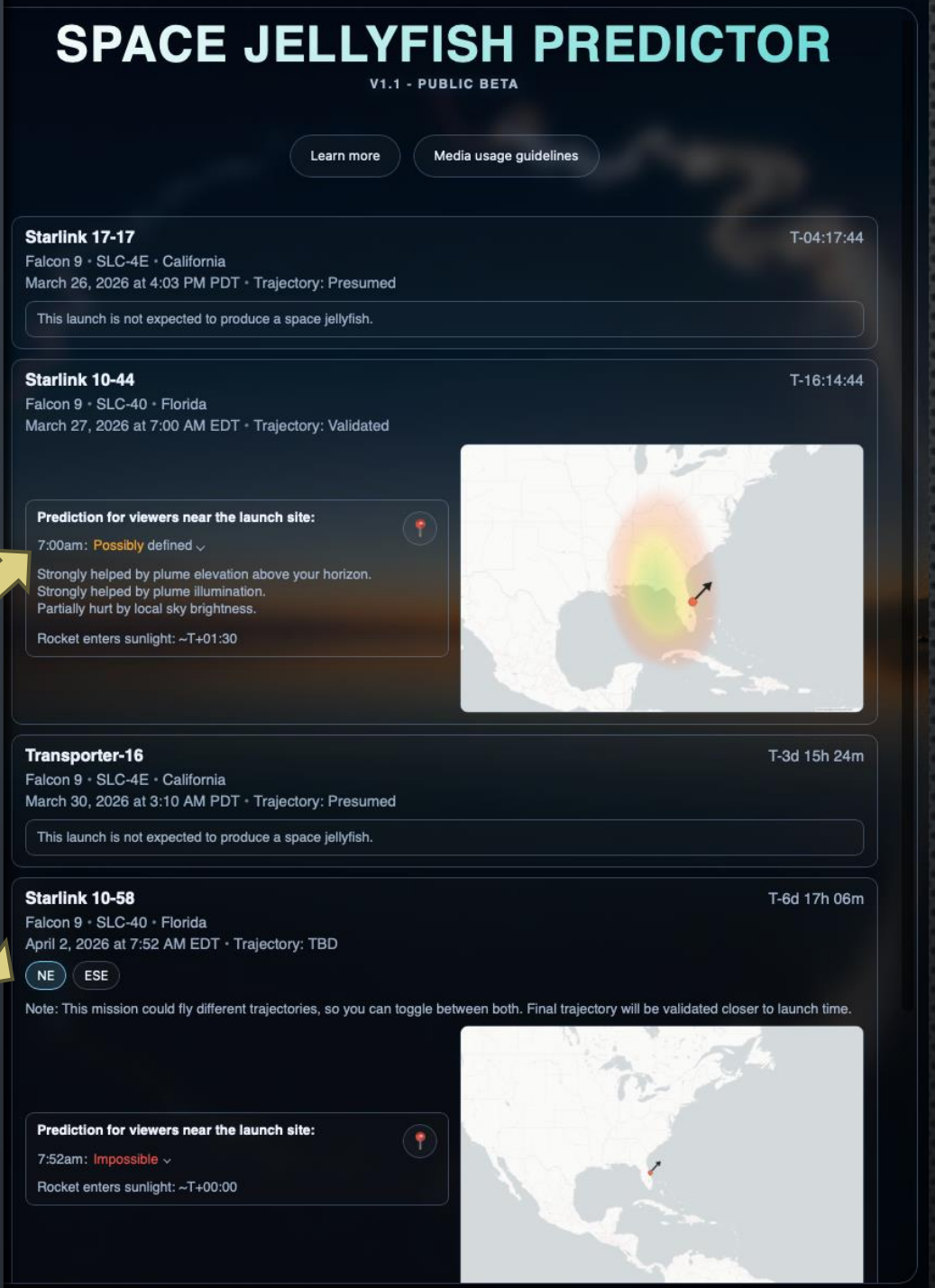
- SHOW TIMES IN THE TIMEZONE LOCAL TO THE LAUNCHPAD AND LOCAL TO THE OBSERVER
- MODEL HOW A 24 HOUR DELAY MIGHT IMPACT JELLYFISHIFICATION
- ADD THE FORECASTED CLOUD COVER BASED ON FORECAST FROM `HTTPS://API.WEATHER.GOV/GRIDPOINTS/MLB/0/0/FORECAST`

OF COURSE THERE'S AN APP FOR THAT

https://jellyfish.johnkrausphotos.com

still just a forecast, based on limited info

trajectory isn't always well known



WANT MORE SPACEY STUFF?

- GETTING STARTED WITH CODING AND AI
- SOURCECODE FOR THIS AND OTHER SPACEY PROJECTS
- WHERE TO LEARN MORE
- JOIN THE EMAIL LIST

https://rtphokie.github.io